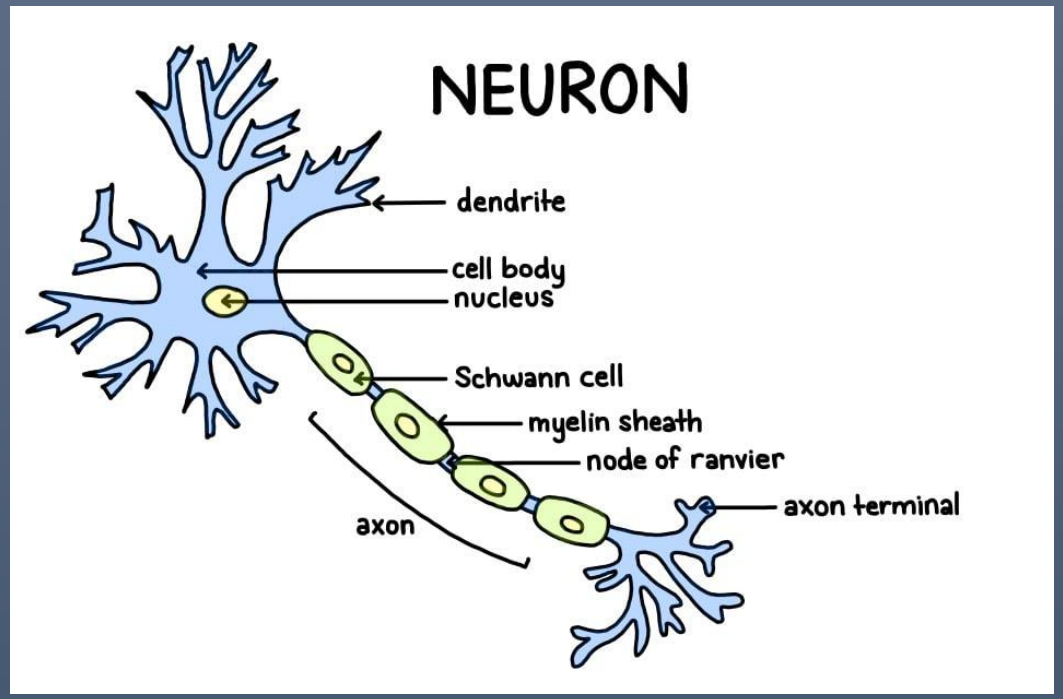
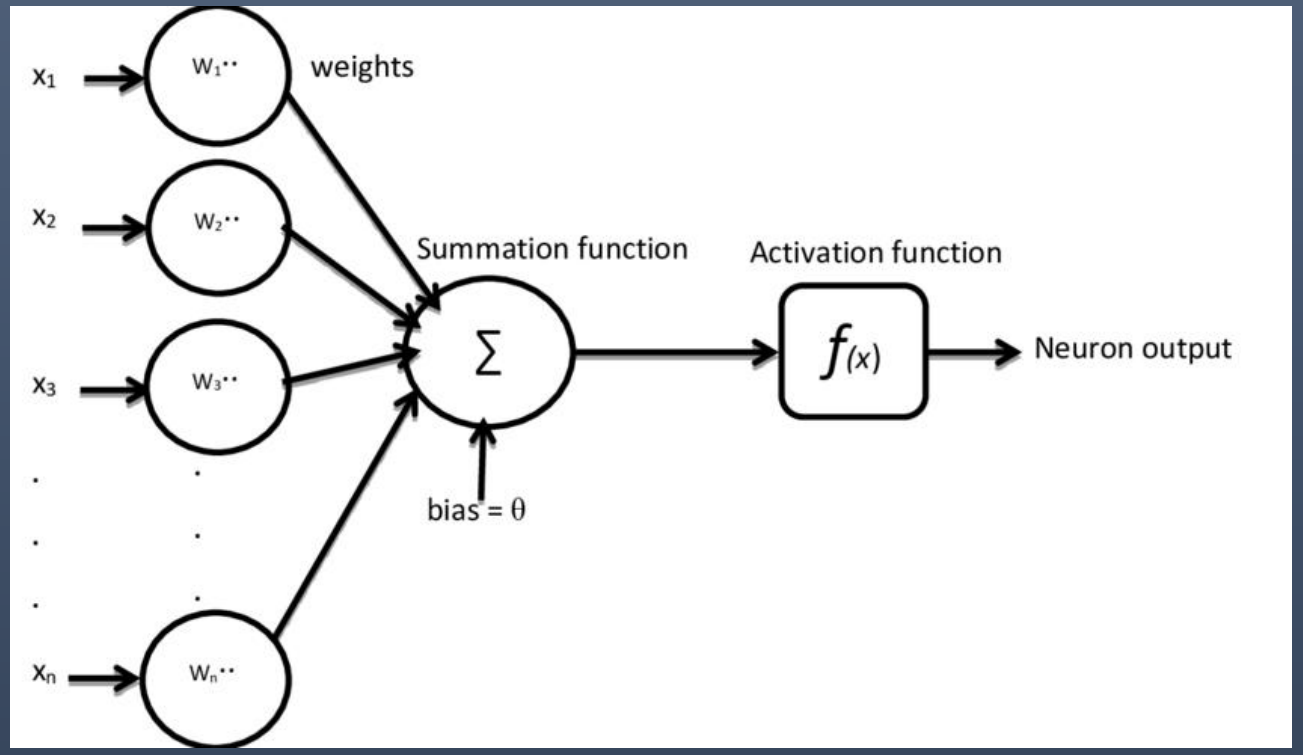
Welcome to Machine Learning




NEURON
dendrite
cell body
nucleus
Schwann cell
myelin sheath
node of ranvier
axon terminal
axon


x1
x2
x3
xn
w1
w2
w3
wn
weights
Summation function
Activation function
Σ
f(x)
Neuron output
bias = θ

What is Machine Learning?

Machine learning is a field of artificial intelligence where computers learn from data using Mathematical Models to

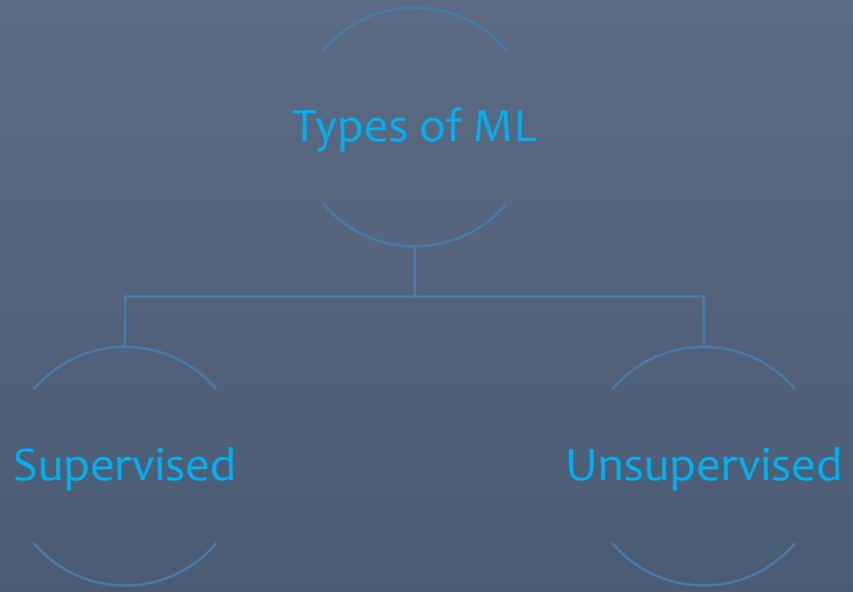
- improve performance on a specific task
- without being explicitly programmed.

It involves developing algorithms(models) that can identify patterns and make predictions or decisions based.

Originating in the 1950s with early efforts in computational models that could learn from data, Arthur Samuel's work on game-playing algorithms marked a pioneering step.

Task Machine Learning can solve

- Email spam detection
- Predicting house prices
- Forecasting stock prices
- Customer segmentation
- Movie recommendations
- Product recommendations
- Sentiment analysis



Designation	Country	Self_Employed	CreditScore	Loan_Status
Manager	France	No	619	Y
Senior Manager	Spain	No	608	N
Manager	France	Yes	502	Y
Senior Manager	France	No	600	N
Manager	Spain	No	850	Y
Manager	Spain	Yes	601	N

CustomerID	Gender	Age	Income	Score
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40

Types of ML

Supervised

Unsupervised

Regression

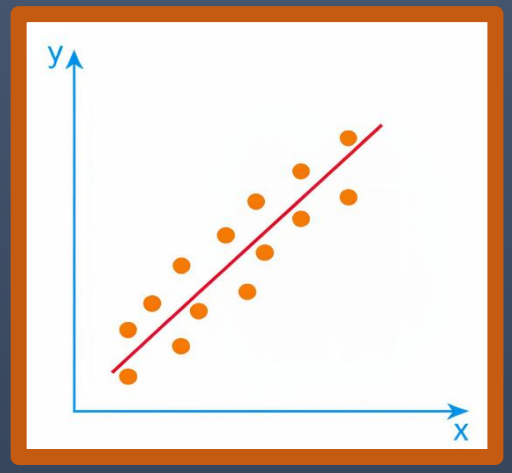
Classification

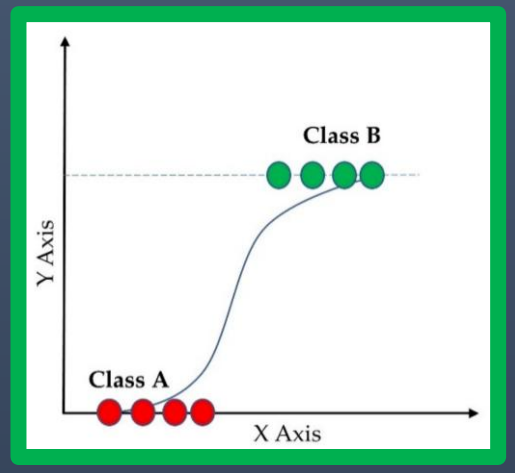
Clustering

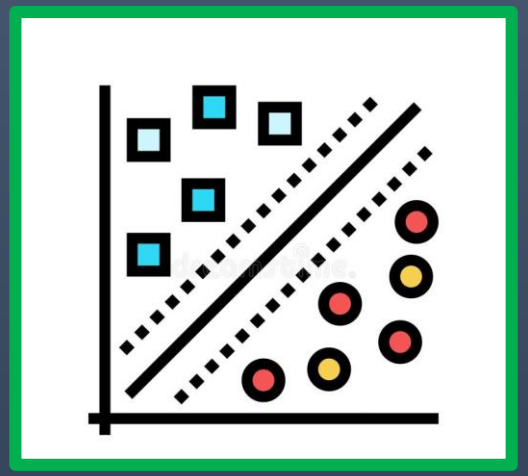
Linear Regression

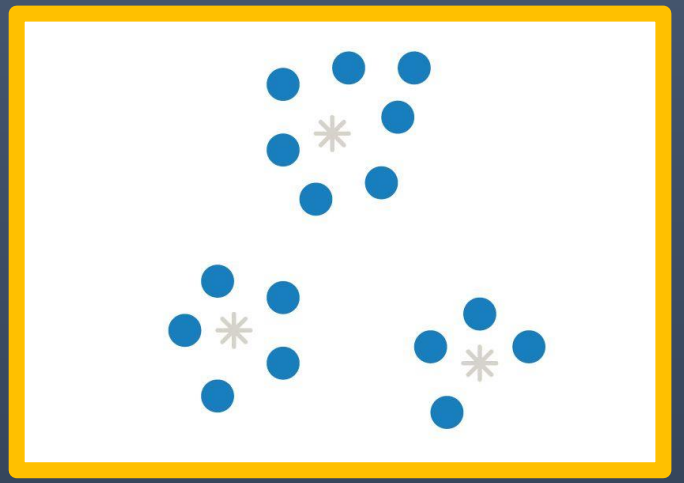
Logistic Regression

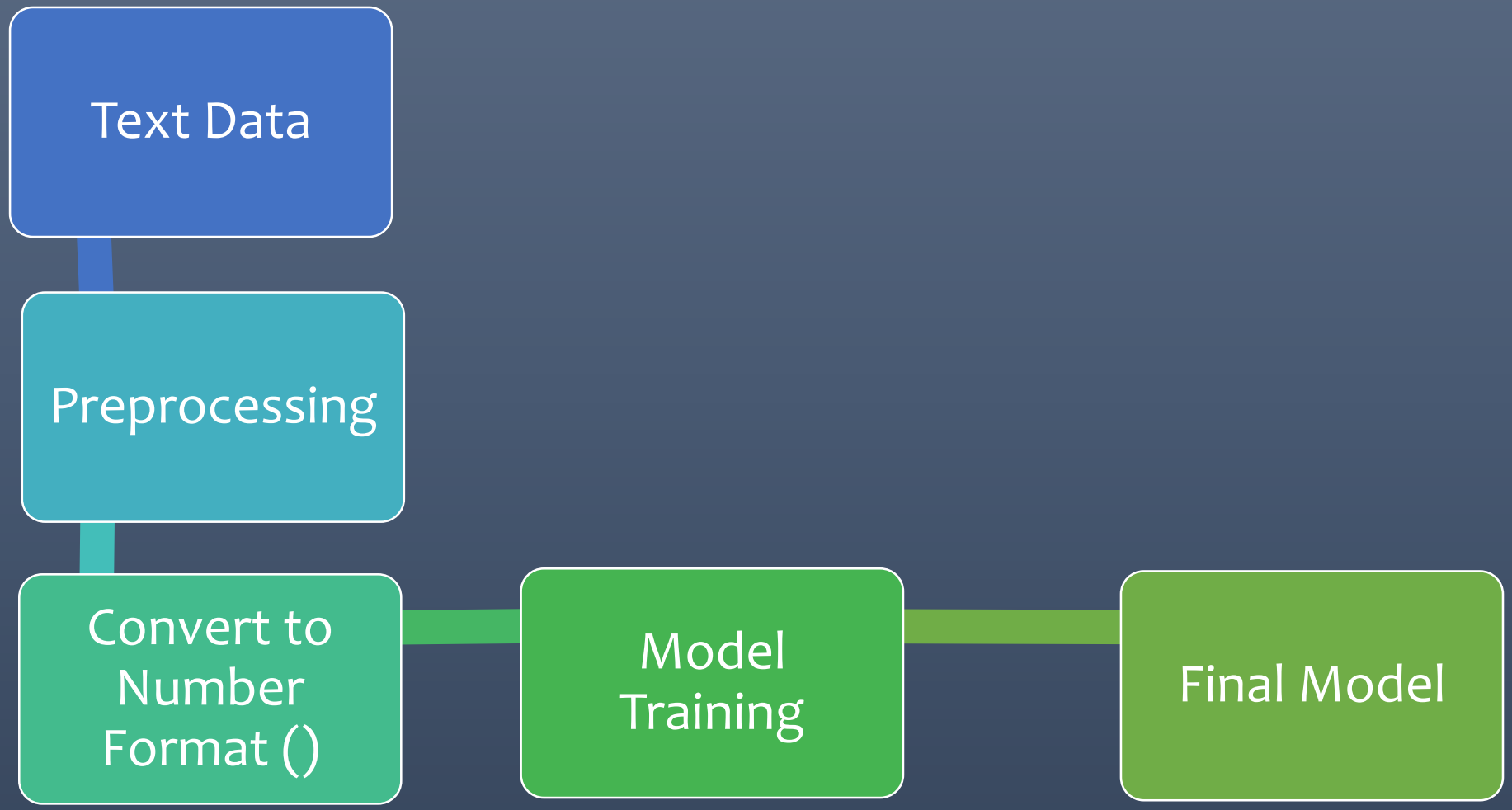
K-Means Clustering










Text Data
Preprocessing
Convert to Number Format ()
Model Training
Final Model

Machine learning Lifecycle (Training)

Designation	Country	Self_Employed	CreditScore	Loan_Status
Manager	France	No	619	Y
Senior Manager	Spain	No	608	N
Manager	France	Yes	502	Y
Senior Manager	France		699	Y
Manager	Spain		850	Y
Manager	Spain	Yes	645	Y
Manager	Spain	Yes	645	Y
Manager	Spain	Yes	645	Y
Manager	France	No		Y
Manager	France	No		N
Executive	France	No	528	Y

Designation	Country	Self_Employed	CreditScore	Loan_Status
Manager	France	No	619	Y
Senior Manager	Spain	No	608	N
Manager	France	Yes	502	Y
Senior Manager	France	No	699	Y
Manager	Spain	No	850	Y
Manager	Spain	Yes	645	Y
Manager	France	No	501	Y
Manager	France	No	684	N
Executive	France	No	528	Y

Pass Cleaned data to the Model to train the Model

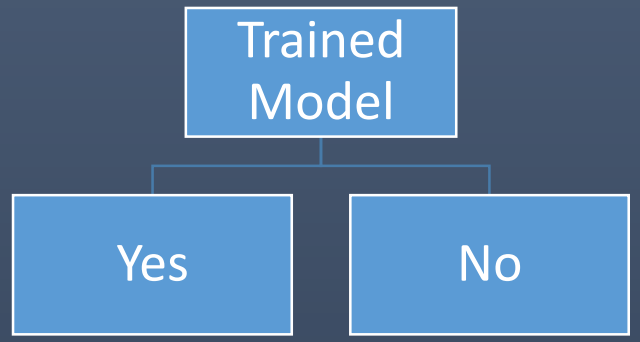
Trained Model

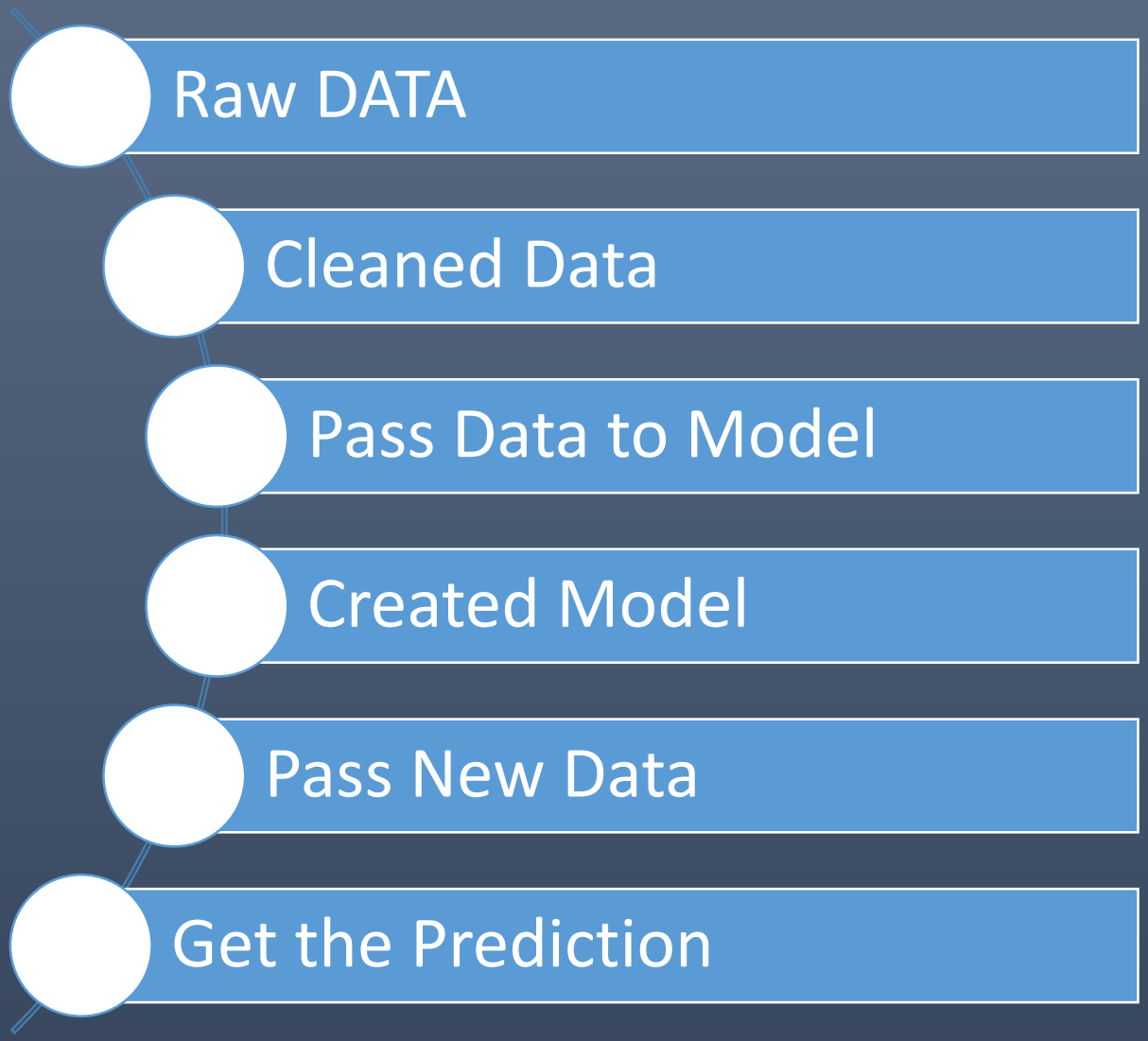
Designation	Country			Self_Employed	CreditScore	Loan_Status
Designation	France	Spain	Germany	Self_Employed	CreditScore	Loan_Status
2	1	0	0	0	619	Y
3	0	1	0	0	608	N
2	1	0	0	1	502	Y
3	1	0	0	0	699	Y
2	0	1	0	0	850	Y
2	0	1	0	1	645	Y
1	1	0	0	0	822	Y
3	0	0	1	0	376	N
2	1	0	0	0	501	Y
2	1	0	0	0	684	N
1	1	0	0	0	528	Y

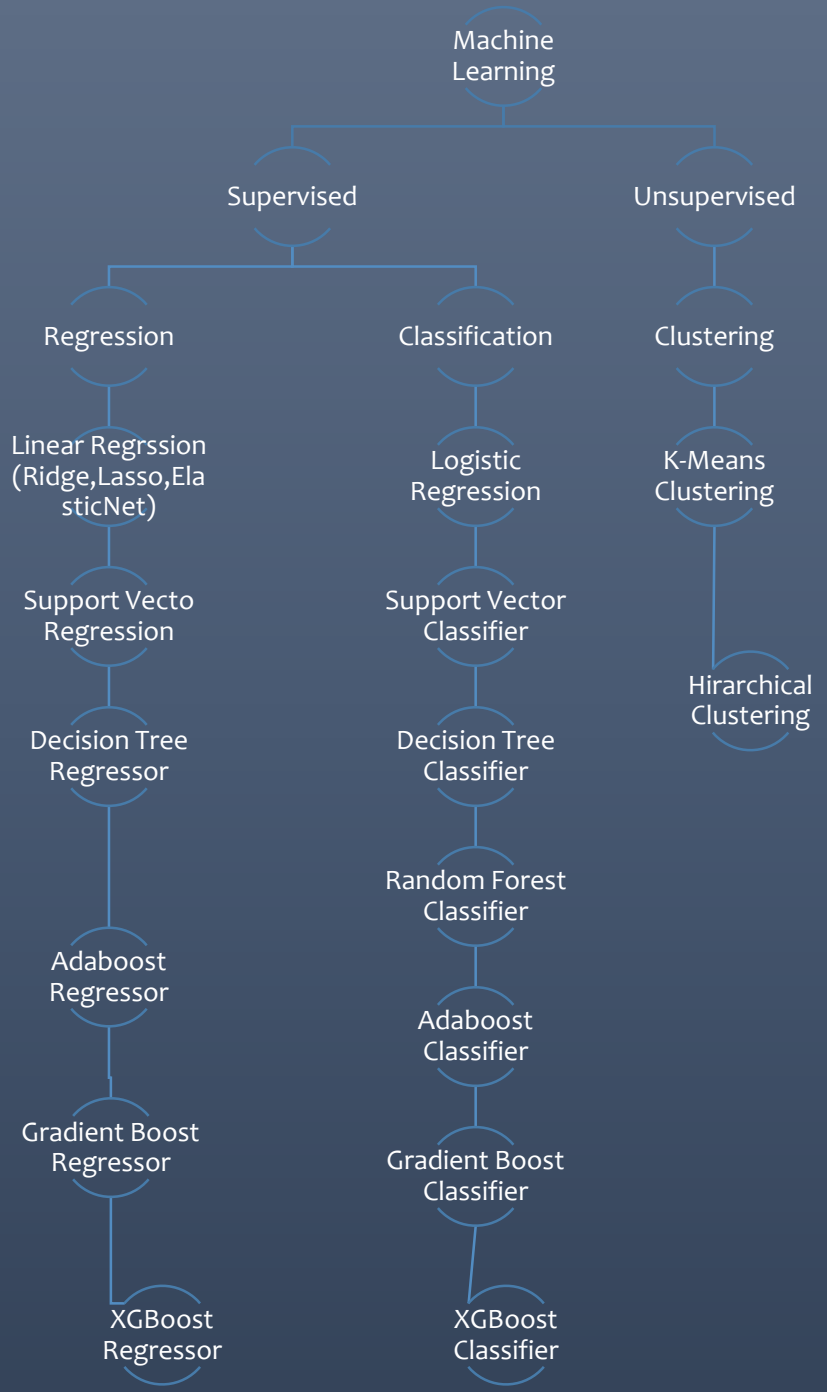
Machine learning Lifecycle (Prediction)

Designation	Country	Self_Employed	CreditScore
Manager	France	No	619

Designation	France	Spain	Germany	Self_Employed	CreditScore
2	1	0	0	0	619




Raw DATA
Cleaned Data
Pass Data to Model
Created Model
Pass New Data
Get the Prediction


Machine Learning
Supervised
Unsupervised
Regression
Classification
Clustering
Linear Regrssion (Ridge,Lasso,ElasticNet)
Logistic Regression
K-Means Clustering
Support Vecto Regression
Support Vector Classifier
Hirarchical Clustering
Decision Tree Regressor
Decision Tree Classifier
Random Forest Classifier
Adaboost Regressor
Adaboost Classifier
Gradient Boost Regressor
Gradient Boost Classifier
XGBoost Regressor
XGBoost Classifier